ສະບາຍດີ!

SABAIDEE! · STUDENT TALK

A Cabin That Keeps Itself Warm

An autonomous heating system built with a Raspberry Bluetooth thermometers and a smart plug

RYU Cheol · ETRI · 6 November 2026

THE PROBLEM

An empty cabin in winter

Pipes can freeze

Nobody is there for months. Water pipes crack when the cabin gets too cold.

Warm on arrival

On a visit the cabin should already be comfortable, without heating all week.

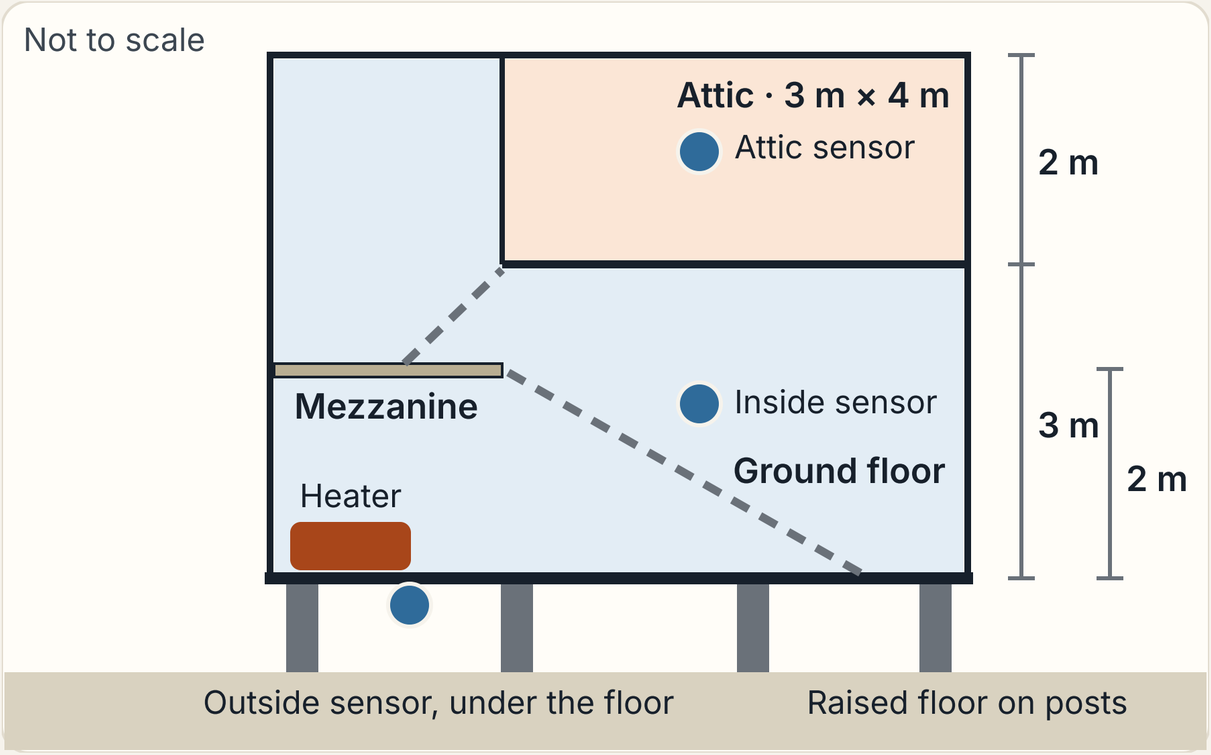
No steady internet

A cloud service fails when the connection drops. The decision must be made on site.

Goal: protect the cabin automatically, and never depend on the internet.

THE CABIN

Three levels, three sensors





Ground floor 3 m high · attic 2 m high · mezzanine 2 m above the floor, open to the ceiling · raised floor on posts

BIG PICTURE

Sense, think, act



Sense Think Act

Two independent programs

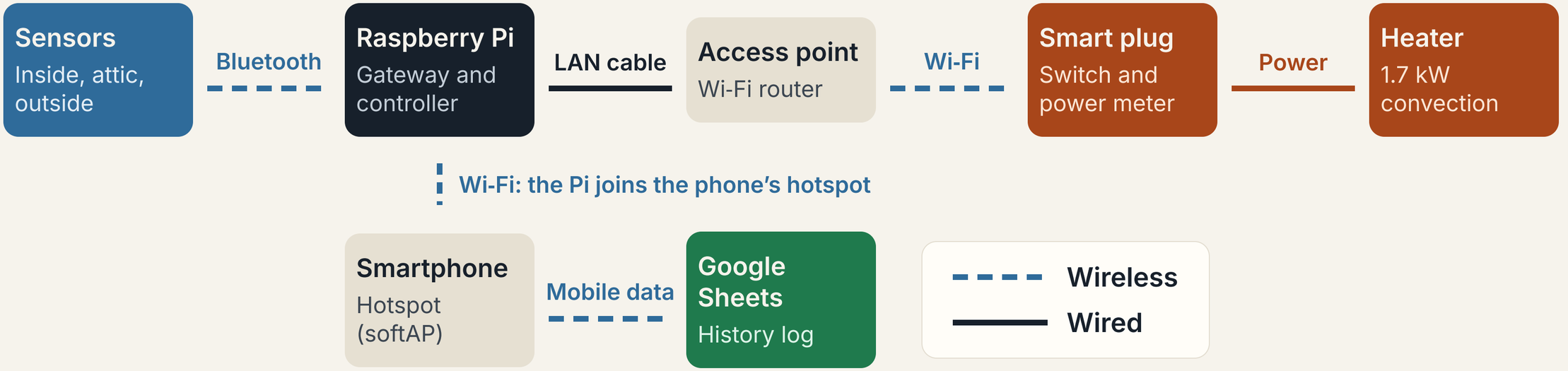
The gateway and the controller only share one file. If one crashes, the other keeps running.

Local first

Control needs no internet. Google Sheets logging is a bonus when a connection exists.

NETWORK

How the devices talk



Control stays local

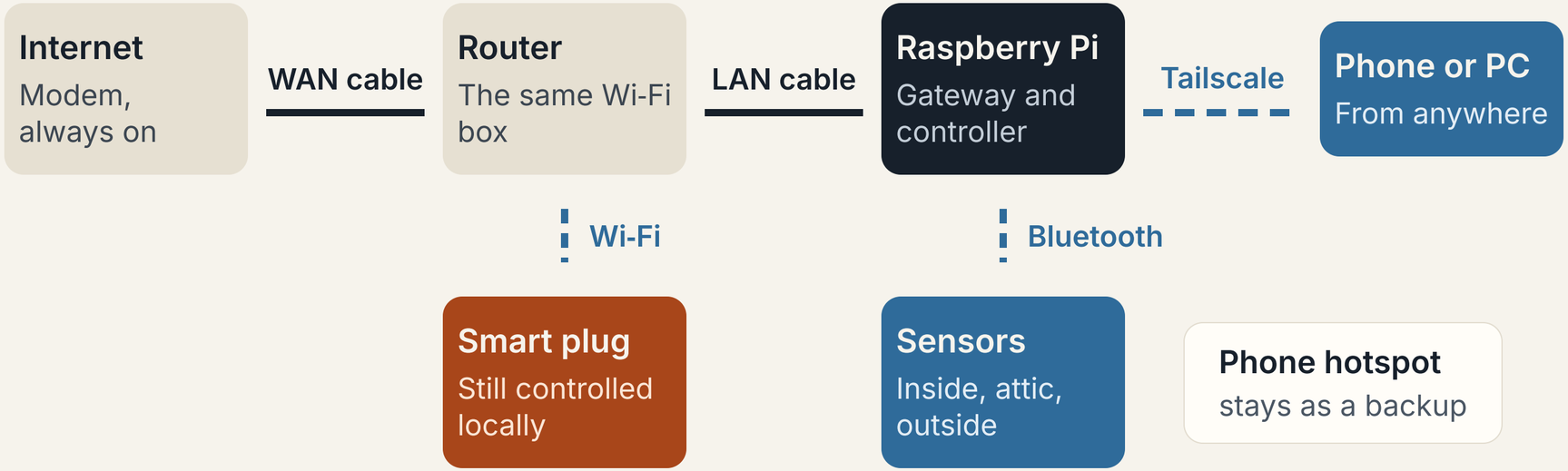
Pi, access point and plug are all inside the cabin network. The heater needs no internet.

History goes out when possible

Without the hotspot the Pi saves readings locally and uploads them later.

NEXT STEP

When a permanent internet line arrives



Always uploading

Readings reach the Sheet even when nobody is there.

Log in from afar

Remote login with no open ports on the router.

Alerts and tuning

Planned: fault alerts, a better heat model and weather forecasts for smarter preheating.

SENSE

Cheap sensors, quietly listening

Xiaomi LYWSD03MMC

A small Bluetooth thermometer and hygrometer, very low cost.

Passive listening

The Pi never connects. It only hears the broadcasts (about every 10 minutes), so batteries last.

Encrypted messages

Each sensor has its own key. The Pi decrypts every broadcast to read the values.

Store and forward

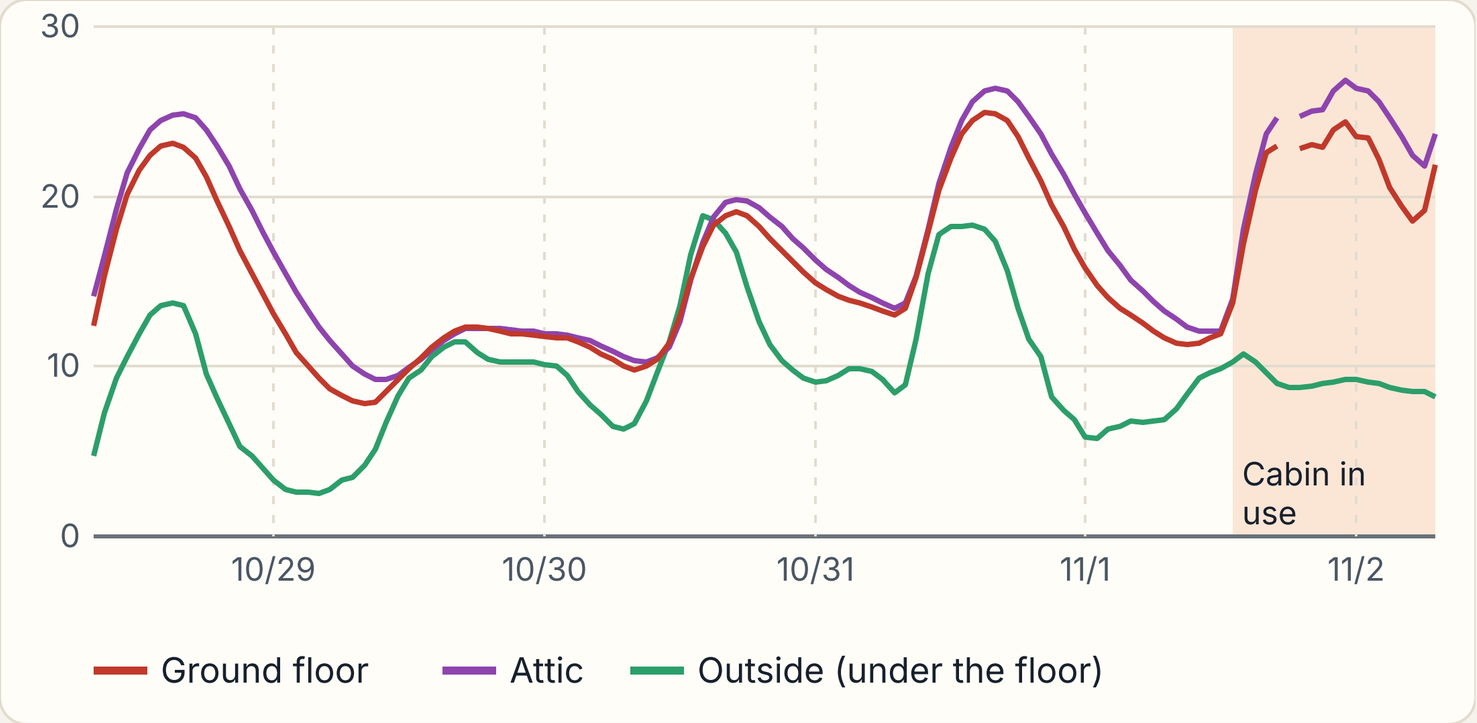
Readings are saved on the Pi and uploaded to Google Sheets when internet is available.

Sensor	Used for
Inside	Frost and comfort control
Outside	Outdoor temperature for preheating
Attic	Sleeping mode and backup sensor

Three sensors, three places, three jobs.

DATA

Every reading lands in a Google Sheet



Hourly temperature, °C · real data

Until the owner arrives (1 Nov, 13:00) the rooms simply follow the weather. Afterwards the heater lifts the ground floor to about 23 °C.

Now	°C	% RH
Attic	25.5	44.2
Outside	8.9	90.8
Ground floor	23.1	45.5

Snapshot from 2 Nov, 08:57. The Pi uploads whenever it has internet.

ACT

A smart plug switches a 1.7 kW heater

1.7 kW

Power of the convection heater when it is ON.

LAN

Commands travel Pi, router, plug. No cloud in the control loop.

3 in 1

Switch, power meter and energy counter in one device.

Why the power meter matters: if the plug is ON but the heater draws almost nothing, the system knows the heater has failed.

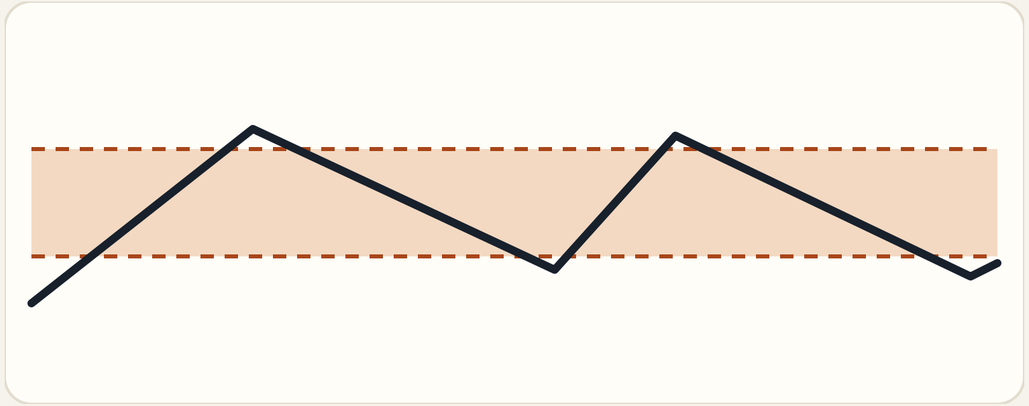
Simple heater: a convection panel with a manual dial. The dial stays at its maximum, so the controller decides when it runs.

THINK

Three modes, one simple rule

- Below the ON value: heater ON
- Above the OFF value: heater OFF
- In between: keep the current state

Why a gap? Without it the relay would click on and off every minute.



Mode	When	Sensor	ON below / OFF above
Frost	Nobody is there	Inside	3 °C / 5 °C
Comfort	You have arrived	Inside	23 °C / 23.5 °C
Sleep	Sleeping in the attic	Attic	20 °C / 21 °C

PLAN AHEAD

Preheating with physics

The owner usually arrives for the weekend at a fixed time: **Friday 20:00**.

Start early by

$$\tau \times \ln\left(\frac{T_{eq} - T_{in}}{T_{eq} - T_{target}}\right) \times \text{margin}$$

$$T_{eq} = T_{out} + \frac{P}{UA}$$

τ = 9.16 h (how slowly the cabin cools) · UA = 47 W/K (heat lost per degree) · P = 1,740 W (heater power) · margin = 1.25

Example: inside 5 °C, target 23 °C

Outside	Start before arrival
8 °C	6.8 hours
0 °C	9.5 hours
−10 °C	10 hours (limit)

1. Estimate

Hours of heating needed, from the outside temperature.

2. Start early

Switch to comfort mode before Friday 20:00 arrival.

3. No show?

No confirmation by midnight: back to frost mode.

SAFETY FIRST

Assume that things will fail

Stale sensor

No fresh reading for 45 minutes? In frost mode the heater turns ON.

Two thermometers

Ground floor and attic back each other up if one sensor dies.

Dead-man timer

While OFF, the plug’s own countdown is renewed every 5 minutes. If the Pi dies, it expires and the plug turns ON.

Time limits

No new command for 36 hours? Comfort and sleep modes fall back to frost mode.

Fault detection

Plug ON but under 100 W for 5 minutes? The heater is probably broken. Log it.

Self-test

Before leaving, one command pulses the plug and checks the power: PASS or FAIL.

PROVE IT

Tests and simulation

57 automated tests

The decision logic is pure functions, so it runs on a laptop. No heater needed.

A real bug, found by a test

The “no-show” check was inside a Friday-only block but waited for Saturday. It could never run. Writing the test exposed it.

Simulator: time within ±0.5 °C

89% → 74%

Same controller, sensor delay 0 vs 5 minutes. A slow sensor makes good control harder.

Simulated room with assumed parameters. People notice changes of about 0.4 °C (Battistel et al., 2023).

YOUR TURN

What would you build in Laos?

Cooling

Fans or air conditioners that run only when a room is too hot.

Greenhouse

Keep temperature and humidity in range, even when the network is down.

Storage rooms

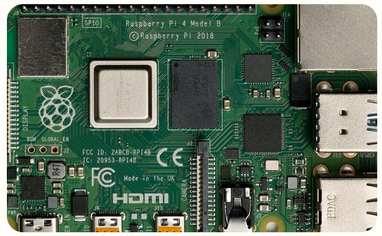
Protect rice, coffee or seeds from heat and damp.

Discussion: when your system fails, what is the safe state, ON or OFF?

Code and notes: github.com/ryucheol/sc-sensors

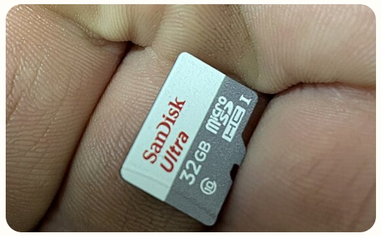
WHAT IT TAKES

Parts and budget



Raspberry Pi 4

$55–80 est.



microSD 32 GB

$5–10 est.



Thermometer × 3

₩17,544 · $12.5



Wi-Fi smart plug

₩4,650 · $3.3



Electric heater

₩59,900 · $42.8

Total of the five items

$119–149

Low: 55 + 5 + 12.5 + 3.3 + 42.8 ≈ 119

High: 80 + 10 + 12.5 + 3.3 + 42.8 ≈ 149

Questions?

Real prices: thermometer ₩5,848 each, plug ₩4,650 each, heater ₩59,900 (US$1 ≈ ₩1,400). Pi and microSD are estimates from AliExpress listings, Nov 2026; no power supply or shipping. Photos: Wikimedia Commons, Laserlicht and J. Halicki (CC BY-SA 4.0), plug and microSD (CC0); heater: product listing image.